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Magnetostratigrahy of the Exposed Lava Section East of the IRDP Drill Hole in Reydarfjordur, Eastern Iceland

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Magnetic polarity directions (VGP) of 154 exposed Miocene basalt lavas from Reydarfjordur, eastern Iceland, can be used successfully for correlation with the IRDP drill core. Three partly synchronous lava sections east of the drill site are predominantly normally magnetized and are stratigraphically of the same age as lavas dated (K-Ar) at about 9 to 10.5 m.y. (*McDougall et al.*, 1976). The age range and magnetic properties of these lavas, the landward equivalents to marine magnetic anomaly 5 (epoch 9), suggest that several reversely magnetized lavas in one of the investigated sections were remagnetized by a 10 m thick dyke. The data presented suggest that in fresh lavas with low Curie temperatures, relatively thick dykes (about 10 m) may cause thermal overprinting of the host rocks to a distance from the dyke equal to its thickness. These results are in contrast with the generally accepted thesis that dykes affect host rocks only up to about 1/5 of their thickness. No systematic decrease in magnetic intensity with depth was observed for a 0.9 km thick vertical lava section in Holmatindur, about 13.5 km east of the drill site. It is suggested that the heterogeneity of the lava pile in terms of magnetic intensity reflects primarily the distinct geochemistry and/or grain size of magnetic phases of individual lavas.

INTRODUCTION

The paleomagnetic history of the exposed Tertiary basalt lava succession in eastern Iceland has been correlated with the geomagnetic time scale [e.g., *Watkins and Walker*, 1977; *Dagley et al.*, 1967]. Lavas in the mountainous Reydarfjordur area (Figure 1) dip gently (2°-12°) toward the neovolcanic zone west of the research area. Here, a 1919 m deep hole was drilled in 1978 [*Scientific Party*, 1979] (Iceland Research Drilling Project, henceforth IRDP), with a 99.7% core recovery. This drilling operation was the first continuous coring of its kind in Iceland. The cliff section east of the drill site, typically 1000 m high, includes lava formations that are intersected at depth below sea level. In accordance with models of crustal generation, the lava formations in the cliff section, although thinner, should have the same paleomagnetic signature as the lavas in the cored sequence. Lithological and magnetic correlations of the cored and exposed lava sequences were regarded as fundamental to an interpretation of the three-dimensional structure of the lava succession. For studying time-dependent variations in the stratigraphic succession it is essential to assign an age to the rock formations. Provided that two objectives are met, the task of dating the IRDP core can clearly benefit from available combined paleomagnetic and K-Ar dating work from the surrounding region. These objectives are (1) magnetic correlation of lavas east of the drill site with paleomagnetic profiles about 20 km to the north in the strike direction, where lavas had been dated by the K-Ar method [*McDougall et al.*, 1976], and (2) lithological correlation of the exposed lavas east of the drill site with the cored lava sequence.

It was anticipated that the present paleomagnetic investigation, substantiated by a detailed stratigraphic study [*Helgason and Zentilli*, this issue], would help constrain the three-dimensional structure and mode of formation of a 3.5 km thick vertical crustal transect. The investigated transect, generated at an accreting plate boundary, extends for about 20 km in the dip direction toward a paleo-rift zone. For this reason a 1 km thick vertical lava section updip of the drill hole was sampled for paleomagnetic studies in Holmatindur in 1979 (section RF, Figure 2). However, when correlation was attempted with earlier paleomagnetic work in the region, the paleomagnetic record of the Holmatindur lavas proved unexpectedly complicated. Despite lithological correlation of lavas in Holmataindur and section E, about 20 km north of the drill site [*Watkins and Walker*, 1977], the study in 1979 indicated contradicting magnetic correlation for these two sections. The Holmatindur section had frequent magnetic reversals whereas section E contained a long sequence of normally magnetized lavas that have been correlated with 'seafloor magnetic anomaly 5,' epoch 9, henceforth referred to as anomaly 5. Further sampling in the area was essential in order to solve this conflict, and in 1980, two additional sections were sampled between Holmatindur and the drill site, section RD (Kollur) and section RE (Teigargerdistindur) on Figure 2. This disagreement has now been resolved, and the author contends that it is due to the local effects of dykes (see below).

DRILL SITE LOCATION AND REGIONAL GEOLOGY

Three major extinct volcanic centers surround the western part of Reydarfjordur (Figure 1). The IRDP drill site is located in a basalt lava sequence only 8 km east of the core of the Thingmuli

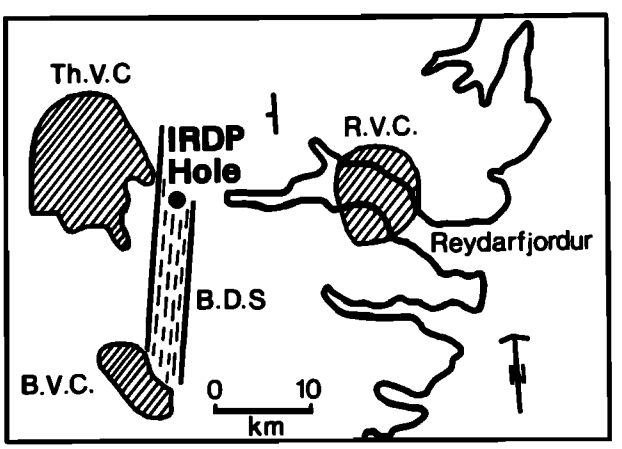


Fig. 1. Location of three volcanic centers in the Reydarfjordur area, i.e., Thingmuli (Th.V.C.), Breiddalur (B.V.C.), and Reydarfjordur (R.V.C.). The IRDP drill hole is located within the Breiddalur dyke swarm (B.D.S.).

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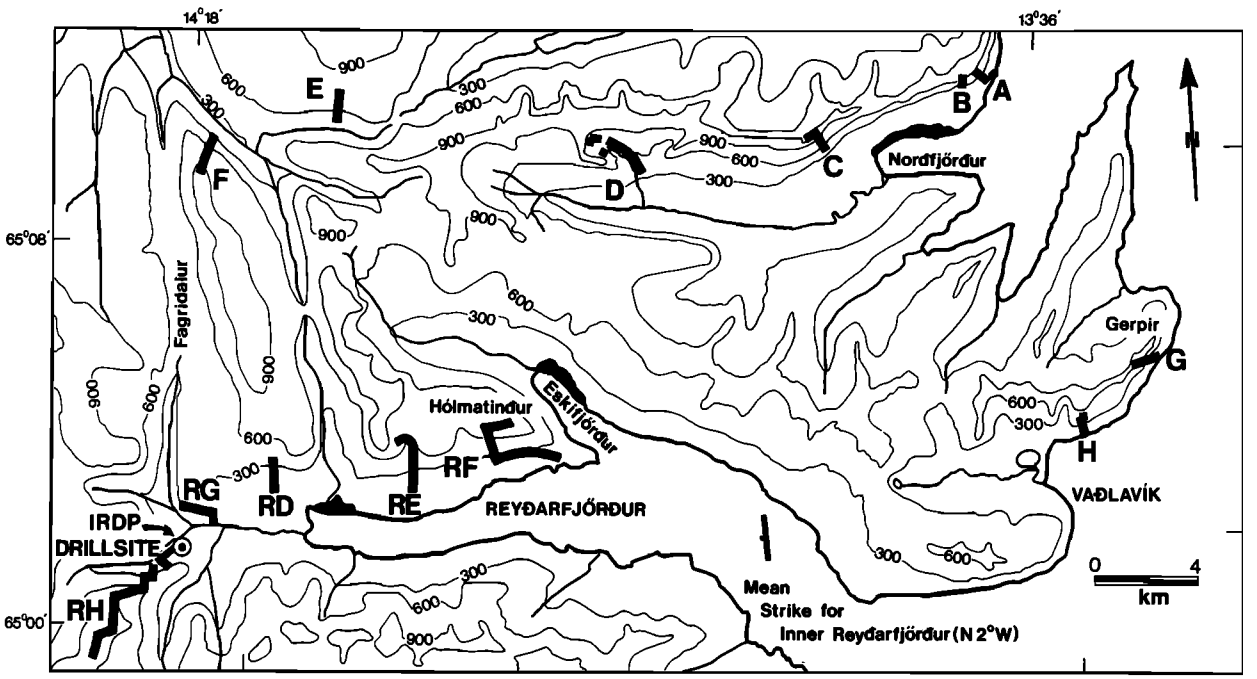


Fig. 2. Location of investigated profiles in Reydarfjordur: Kollur (RD), Teigargerdistindur (RE), Holmatindur (RF), Graenafell dyke section (RG), Trollafjall lava/dyke section (RH), and previously studied sections in the region [*Watkins and Walker*, 1977] : G, H, A, B, C, D, E, and F.

volcanic center. A dyke swarm intersecting the lava sequence at the drill site (10% crustal dilation) has been traced to Breiddalur volcanic center (v.c.) approximately 20 km to the south [*Walker*, 1963]. In a study of the Breiddalur dyke swarm, in Graenafell, 1 km north of the drill site (section RG on Figure 2), the mean dyke thickness for 51 dykes was 3.7 m and only 6% exceeded 10 m. The first exposed units from the Breiddalur v. c. were erupted approximately 9 m.y. ago [*McDougall et al.*, 1976; *Watkins and Walker*, 1977]. All quoted dates by *McDougall* in this paper have been recalculated, applying recent K-Ar decay constants [*Steiger and Jaeger*, 1977]. Field relationships suggest that although activity in Thingmuli v. c. overlapped in time the Breiddalur v. c., volcanism

TABLE 1. Summarized Information on Sections Sampled for This Study

Section*	Formation	Number of Units	Number of Mini-cores per Unit	Magnetic Polarity
Kollur (RD)	Kollur porphyritic basalt	35	4	N
	Tholeiite - 5			N
	Kambfell porphyritic basalt			N
Teigargerdistindur (RE)	Kollur porphyritic basalt	29	4	N
	Tholeiite - 5			N
	Kambfell porphyritic basalt			N
	Tholeiite - 4			N
	Reydarfjordur porphyritic basalt			N
Holmatindur (RF)	Kollur porphyritic basalt	90	4 (or less)	Mainly normal with eight short reversed polarity zones
	Tholeiite - 5			
	Tholeiite - 4			
	Tholeiite - 3			
	Ljosa olivine basalt			
	Tholeiite - 2			R - N - R
	Grjota olivine basalt			
	Tholeiite - 1			R
	Holmar olivine basalt			R

For further details, see *Helgason and Zentilli* [this issue].
* See location on Figure 2.

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TABLE 2. Paleomagnetic Results for Sections RD (Kollur), RE (Teigargerdistindur), and RF (Holmatindur)

NO	N	P	Ī	D	R	α_{95}	θ'	φ'
RD3	4	N	73.9	59.2	3.9942	4.0	63.2	58.5
RD4	4	N	80.7	41.4	3.9956	3.5	73.7	33.1
RD5	4	N	83.6	40.5	3.9986	1.9	72.7	14.2
RD6	4	N	66.1	81.9	3.9809	7.3	45.9	56.3
RD7	4(1)	N(T)	59.1	94.6	3.9980	2.3	33.7	52.7
RD8	4	N	67.3	94.5	3.9941	4.0	42.3	45.8
RD9	3	N	67.6	89.5	2.9965	5.1	44.5	49.0
RD11	4	N	67.8	93.2	3.9888	5.6	43.4	46.2
RD13	4	N	64.1	66.1	3.9974	2.7	50.4	71.5
RD14	4	N	64.8	55.2	3.9914	4.9	55.7	80.5
RD15	4	N	71.5	28.7	3.9735	8.6	73.7	94.5
RD16	4	N	63.8	70.7	3.9954	3.6	48.0	68.0
RD17	3(1)	N	63.2	44.2	2.9810	9.6	58.5	94.1
RD18	4	N	71.3	354.6	3.9909	5.0	80.6	183.9
RD19	3(1)	N	75.3	19.3	2.9878	9.6	81.1	84.0
RD20	3(1)	N	64.6	8.9	2.9896	8.9	70.8	146.9
RD21	4	N	77.0	76.6	3.9354	13.6	59.7	39.9
RD22	4	N	71.9	73.5	3.9744	8.5	55.5	53.8
RD23	4	N	64.2	25.9	3.9735	8.6	66.4	116.6
RD24	4	N	63.7	36.5	3.9863	6.2	62.0	102.6
RD25	4	N	57.4	30.5	3.9987	1.9	57.7	117.4
RD26	4	N	73.0	358.5	3.9278	14.4	83.5	172.6
RD27	4	N	65.3	69.2	3.9938	4.1	50.2	67.4
RD28	4	N	57.5	63.7	2.9902	8.6	45.0	80.3
RD29	4	N	59.7	52.2	3.9955	3.5	51.8	89.5
RD30	4	N(T)	56.3	76.3	3.8182	23.3	38.6	69.7
RD31	4	N(T)	53.3	66.9	3.9654	9.9	39.9	80.3
RD32	4	N	60.5	16.7	3.9884	5.7	64.6	135.6
RD33	4	N	50.8	9.1	3.9522	11.6	56.0	151.8
RD34	4	N(T)	44.1	245.2	3.9801	7.5	13.7	288.6
RD35	4	N	70.4	17.7	3.9809	7.3	76.4	117.2
RD35A	4	N	63.0	28.7	3.9886	5.6	64.1	114.2
RD36	4	N(T)	60.6	157.5	3.9843	6.6	18.0	3.3
RD37	4	N	74.2	98.4	3.9981	2.3	49.4	34.2
RD38	4	N	73.8	50.4	3.9782	7.8	66.8	9,8
RE6	4	N	72.7	45.7	3.8516	20.9	67.7	72.3
RE7	4	N	64.6	50.9	3.9822	7.0	57.2	85.0
RE8	4	N	67.5	2.1	3.9924	4.6	75.3	160.6
RE9A	4	N	61.7	33.2	3.9990	1.6	61.1	109.5
RE9	4	N	77.7	32.2	3.9891	5.5	76.8	54.9
RE10	4	N	58.3	33.3	3.9947	3.8	57.6	112.9
RE11	4	N	67.6	30.3	3.9789	7.7	68.6	104.0
RE12	2(2)	N	60.3	12.0	1.9965	(14.8)	65.3	143.8
RE13	4	N	76.2	25.2	3.9986	1.9	79.1	70.7
RE14	3(1)	N	71.0	1.3	2.9953	5.9	80.4	161.3
RE15	4	N	62.0	359.9	3.9903	5.2	68.2	165.9
RE17	4	N	67.4	21.0	3.9990	1.6	71.6	119.1
RE18	4	N	84.1	312.0	3.9967	3.0	70.9	318.4
RE19	4	N	62.3	27.2	3.9972	2.8	63.8	117.3
RE20	4	N	83.3	29.9	3.9969	2.9	75.0	11.8
RE21	4	N	82.4	68.1	3.9972	2.8	66.4	22.6
RE22	4	N	67.7	62.2	3.9942	4.0	55.6	69.9
RE23	4	N	74.4	51.5	3.9969	2.9	66.9	62.0
RE24	4	N	70.5	35.8	3.9838	6.7	69.7	89.1
RE25	4	N	77.1	21.2	3.9955	3.5	81.1	63.6
RE26	4	N	75.4	23.8	3.9953	3.6	79.2	78.8
RE28	4	N	74.4	32.3	3.9955	3.5	74.9	76.3
RE30	4	N	66.3	58.2	3.9937	4.2	55.9	75.5
RE31	4	N	61.7	66.4	3.9907	5.1	47.8	73.9
RE32	4	N(T)	54.1	254.7	3.9958	3.4	25.0	284.7
RE33	4	N(T)	46.4	248.3	3.9975	2.6	16.4	286.7
RE34	4	N	78.4	296.5	3.9947	3.8	65.5	290.5
RE35	1	N(T)	56.9	154.8			14.3	6.2
RE36	4	N(T)	45.7	286.9	3.9437	12.7	31.5	253.0
RF1	4	R	-64.6	253.4	3.9925	4.5	-47.8	245.0
RF2	4	R	-58.9	257.6	3.9981	2.3	-40.4	246.9
RF3	4	R	-61.1	252.6	3.9979	2.4	-44.6	249.3
RF4	3(1)	R	-61.2	259.8	2.9731	14.3	-41.7	243.2
RF5	4	R	-59.7	229.4	3.9769	8.0	-53.0	272.8
RF6	4	R	-52.1	225.1	3.9957	3.4	-47.8	283.5
RF7	4	R	-72.9	188.9	3.9267	14.5	-57.7	174.7
RF8	4	R	-67.8	188.8	3.9234	14.8	-75.1	324.0
RF9	3	R	-77.2	196.6	2.9869	10.0	-83.1	243.5
RF10	4	R	-65.3	202.5	3.9059	16.5	-68.6	300.7
RF11	4	R	-72.0	210.4	3.9974	2.7	-73.5	270.6
RF12	4	R	-71.8	218.4	3.9912	4.9	-69.9	262.4
RF13	4	R	-56.7	204.8	3.9161	15.5	-58.7	306.1
RF14	3(1)	R	-67.0	222.2	2.9916	8.0	-63.3	270.5
RF15	4	R	-66.2	216.3	3.9984	2.1	-64.8	279.0
RF16	3(1)	R	-51.7	217.3	2.9654	16.3	-50.3	292.8
RF17	3	R	-52.5	211.9	2.9864	10.2	-52.7	299.1
RF19	4	R	-51.1	229.7	3.9918	4.8	-45.2	279.1
RF20	4	R	-54.5	225.5	3.9182	15.3	-49.7	281.4
RF21	4	R	-62.3	222.7	3.9888	5.6	-58.2	277.2

TABLE 2. (Continued)

NO	N	P	I	D	R	α_{95}	θ'	φ'
RF22	3(1)	R	-68.7	191.1	2.9801	12.3	-75.9	317.0
RF23	4	R	-81.0	202.4	3.9953	3.6	-79.1	203.5
RF24	4	R	-78.9	217.7	3.9796	7.6	-75.0	225.5
RF25	4	R	-75.2	225.3	3.9995	1.1	-70.1	243.1
RF26	3	R	-73.4	259.7	2.9957	5.7	-54.8	226.9
RF27	4	R	-73.3	254.3	3.9856	6.3	-56.7	230.5
RF28	3(1)	R	-80.9	246.0	2.9776	13.1	-66.3	209.8
RF29	2(1)	R	-68.9	231.7	1.9963	15.2	-61.4	257.1
RF30	2	R	-65.8	230.4	1.9609	51.0	-58.7	264.0
RF31	2(2)	N(T)	3.8	180.0	1.9998	(3.5)	-23.1	346.0
RF32	4(1)	R(T)	-50.7	304.0	3.9575	11.0	-15.7	213.3
RF33	2(2)	R(T)	-49.9	321.7	1.9935	(20.2)	-10.2	198.8
RF34	4	N(T)	53.0	223.5	3.9069	16.4	14.2	309.7
RF35	3	N	60.3	75.2	2.9876	9.7	42.7	67.8
RF36	4	N	73.9	51.7	3.9895	5.4	66.4	63.7
RF37	4	N	67.7	24.9	3.8979	17.2	70.7	112.2
RF38	3(1)	N(T)	23.2	77.2	2.9961	5.4	16.4	82.5
RF39	4	R(T)	-6.8	130.2	3.9572	11.0	-19.0	39.8
RF40	4	R	-62.4	221.2	3.9940	4.1	-58.9	279.0
RF41	3(1)	R	-62.6	210.0	2.8396	36.4	-62.9	293.9
RF42	4	R	-71.0	211.4	3.9626	10.3	-72.0	273.1
RF43	4	R	-77.8	219.7	3.9842	6.6	-73.9	232.3
RF44	4	R	-51.6	177.2	3.9428	12.8	-57.2	350.4
RF45	4	R	-46.2	195.9	3.9955	3.5	-51.2	323.2
RF46	4	R	-43.0	197.8	3.9768	8.1	-48.4	321.4
RF47	4	R	-68.9	204.9	3.9882	5.7	-72.1	289.1
RF47A	4	N	82.9	170.9	3.9762	8.2	51.2	349.5
RF48	4	N	82.8	260.5	3.9946	3.9	59.5	317.5
RF49	4	N	46.6	313.0	3.9874	5.9	42.7	227.6
RF50	4	N	61.9	78.3	3.9931	4.4	43.0	63.8
RF51	4	N(T)	56.7	79.8	3.9918	4.8	37.5	66.7
RF52	4	R	-78.2	223.9	3.9978	1.9	-72.5	228.6
RF53	3(1)	R	-80.2	199.1	2.9995	1.9	-80.8	207.6
RF54	4	R(T)	-23.7	142.3	3.9526	11.6	-31.4	30.4
RF55	4	N(T)	60.2	93.9	3.9411	12.9	35.0	52.7
RF56	4	N	40.2	4.7	3.9811	7.3	47.8	159.5
RF57	2(2)	R	-79.3	199.3	1.9977	(12.0)	-81.4	217.5
RF59	4	R(T)	-65.2	295.8	3.9689	9.4	-32.7	212.6
RF60	4	N	65.1	53.5	3.9878	5.8	56.7	81.9
RF61	4	N	63.5	65.4	3.9881	5.7	49.9	73.1
RF62	4	N	64.3	58.9	3.9971	2.8	53.5	78.0
RF63	3(1)	N	52.9	23.3	2.9959	5.5	55.4	130.4
RF64	2(2)	N	81.2	230.4	1.9969	(13.9)	51.9	324.4
RF65	4	N	78.8	83.5	3.9977	2.5	59.4	31.8
RF66	3(1)	N	64.7	24.6	2.9937	6.9	67.3	118.2
RF67	3(1)	R	-87.0	212.3	2.9974	4.4	-69.8	175.3
RF68	2(2)	R	-83.7	202.0	1.9987	(9.0)	-75.8	185.2
RF69	2(2)	R	-78.6	189.2	1.9992	(7.0)	-85.2	212.1
RF70	2(2)	R	-67.6	258.0	1.9483	(60.2)	-49.0	237.6
RF71	3(1)	N	69.3	356.0	2.9900	8.7	77.7	177.4
RF72	4	N	76.3	346.6	3.9925	4.5	84.2	253.1
RF73	2(1)	N	59.7	30.8	1.9958	16.2	59.9	115.1
RF74	2(2)	R	-79.2	108.2	1.9695	(45.1)	-63.4	117.0
RF75	4	N	61.7	28.0	3.9880	5.8	62.9	116.9
RF76	2(2)	N	56.8	68.2	1.9927	(21.5)	42.4	76.8
RF77	4	R	-82.8	166.8	3.9974	2.7	-78.4	149.9
RF78	4	N	60.8	43.8	1.9957	16.4	56.2	97.8
RF79	4	R	-81.4	38.0	3.9713	9.0	-50.5	149.8
RF80	4	N	62.3	65.4	3.9933	4.3	48.8	74.3
RF81	3	N(T)	57.6	78.8	2.9747	13.9	38.7	66.9
RF82	4	N(T)	48.0	79.1	3.9155	15.6	30.6	72.3
RF83	3	R	-74.4	244.4	2.9742	14.0	-61.7	233.9
RF84	4	N	76.2	59.0	3.9837	6.7	65.5	51.3
RF85	4	N(T)	43.5	267.8	3.9569	11.0	21.9	269.3
RF86	3(1)	N(T)	37.1	247.8	2.9982	3.7	9.9	284.5
RF87	4	N	69.1	67.6	3.9984	2.1	54.9	63.4
RF88	1	N(T)	41.0	286.7			28.2	251.4
RF89	4	R	-66.1	242.9	3.9884	5.7	-53.7	251.9
RF90	4	N(T)	32.0	285.4	3.9922	4.6	22.2	249.4
Dykes								
RF33D	3	R	-62.5	207.8	2.9427	21.1	-63.8	296.4
RF41D	3(1)	N	69.5	7.7	2.8850	30.5	77.6	143.9
RF51D	4	R	-72.1	223.5	3.9963	3.2	-67.9	258.2

NO, unit number; N, number of samples used from each unit; P, magnetic lava polarity, N and R lat > |40°|, N (T) and R (T) lat |10° -40°| (transitional); I,D, tilt corrected magnetic inclination and declination; R, resultant unit vector; α_{95}, radius of a circle with 95% confidence for the mean magnetic direction of each unit; θ' and φ', virtual geomagnetic pole positions calculated after tilt corrections.

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TABLE 3. Tilt Corrections for Section RF (Holmatindur)

Unit Interval	Elevation, m	Tilt Correction
RF 1-47	40-300	+9°
RF 47A-65	300-600	+6°
RF 66-88	600-900	+3°
RF 89-90	>900	+1.5°

in the first continued for some time after the second became dormant [*Walker*, 1963, 1975]. Approximately 20 km east of the drill site is the Reydarfjordur v. c., whose rocks lie stratigraphically far below the lavas at the collar of the drill hole. In this area, volcanic activity ceased approximately 10.5 m.y. ago [*McDougall et al.*, 1976]. It is interesting to note that silicic units typical of the core of Thingmuli v. c. (Figure 1) presently occur at about 1000 m above sea level, whereas corresponding units in the Reydarfjordur v. c. occur close to sea level, reflecting the different degrees of burial that these volcanoes were subjected to. During the drilling experiment it was a matter of speculation whether lavas of the Reydarfjordur v. c. or intrusive phases corresponding to the Thingmuli v. c. would be intersected at depth. No major intrusive phases of Thingmuli v. c. were intersected, but numerous minor intrusives of the Breiddalur dyke swarm occur in the drill core.

PALEOMAGNETIC SAMPLING AND LABORATORY TREATMENT

A total of 154 units were sampled in three sections, RD, RE and RF on Figure 2. Generally, four minicores, 2.5 cm in diameter, were drilled from each lava with a two-stroke portable drill. The samples were oriented in situ with a Brunton compass using both geographic and sun sightings. Each core was sliced, and approximately 8 cm were used for paleomagnetic studies. Information on stratigraphic sections is provided in Table 1. The direction and intensity of natural remanent magnetization (NRM) were measured with a Schonstedt digital spinner magnetometer at Dalhousie University. Applied peak alternating field values during demagnetization ranged from 150 to 250 Oe for sections RD and RE, whereas for section RF the applied field ranged from 25 to 500 Oe in 11 successive steps.

The RF section in Holmatindur was divided into 90 lava units each marked in the field with a painted number. Three intervals of thin flow units occur in the lower part of the section; for each interval only one unit was drilled. Only three segments, each less than 10 m in thickness, are covered with rubble and not exposed in the vertical section. Units 52 through 88 are located in a gully cut by a 10 m thick vertical dyke of reversed polarity (unit RF 51D, Table 2). A total of five thin lavas, i.e., units 35A, 49A, 54A, 54B, and 88A were not sampled for this paleomagnetic study.

PALEOMAGNETIC RESULTS

Results from section RF (Holmatindur)

Following conventional statistical methods [*Fisher*, 1953], the minimum scatter was calculated, i.e., α_{95}, for the mean paleomagnetic direction of each unit. Results for section RF (Holmatindur)

TABLE 4. Tilt Corrections for Sections RD (Kollur) and RE (Teigargerdistindur)

Unit Interval	Elevation, m	Tilt Correction
RD 3-25	0-300	+9°
RD 26-38	>300-600	+6°
RE 6-23	300-600	+6°
RE 24-36	>600-900	+3°

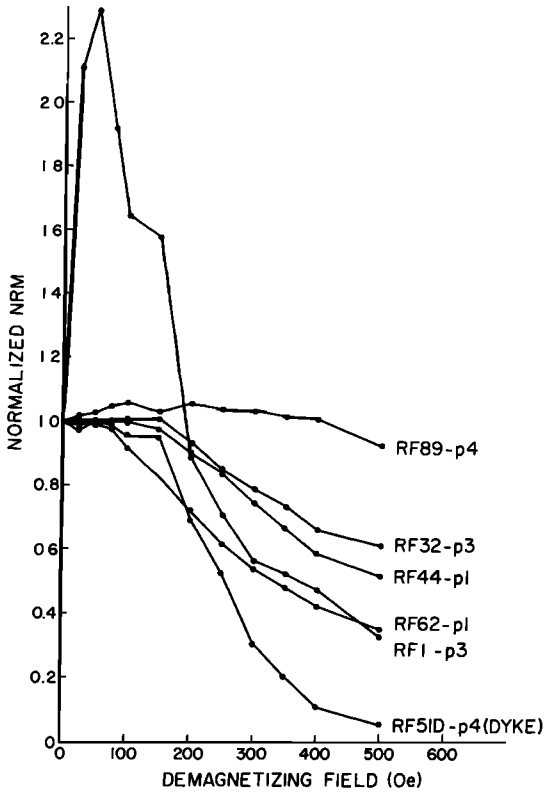


Fig. 3. Normalized magnetic intensity of selected samples from section RF (Holmatindur) plotted against alternating field stength.

are presented in Table 2. For 20 lava units in section RF, $\alpha_{95\Sigma}$ indicated low internal consistency; therefore an additional sample was spun. This usually resulted in excellent agreement for three samples of each lava unit. However, samples were rejected if widely removed from the average unit declination and/or inclination. Of the 342 minicores demagnetized from Holmatindur,

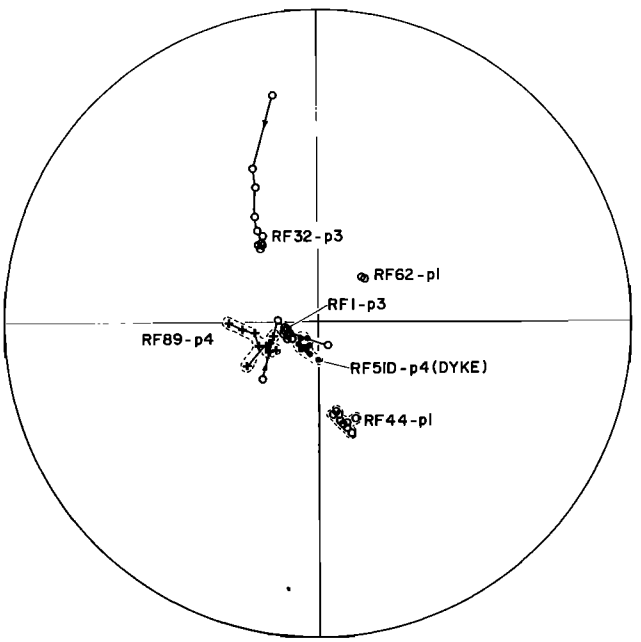


Fig. 4. Stereographic plot of selected samples from section RF (Holmatindur), showing a trend toward steeper and more stable inclinations with increased demagnetizing field stength.

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TABLE 5. Summarized Information on Selected Samples Plotted for Magnetic Properties (Figures 3 and 4)

Sample	Field Classification	Grain Size	Mineral Phases*	Alteration State	Stratigraphic Location** m	Natural Remanence J_0 x 10^{-4} emu/cm³	Magnetic Polarity
RF 1-p3	olivine tholeiite	medium	pl,ol,px,op,alt	chl, minor ol left	70	1.3	R(T)
RF 32-p3	tholeiite	fine	pl,px,op,alt	specks of chl	317	12.8	R(T)
RF 44-p1	olivine tholeiite	coarse	pl,ol,px,op,alt	clay, ol partly altered	426	11.2	R
RF 62-p1	tholeiite	medium	pl,px,op,alt	fresh, minor clay	665	53.4	N
RF 89-p4	tholeiite	fine	pl,px,op	fresh	980	4.7	R
RF 51D-p4	dyke	coarse	pl,px,op,alt	fresh, minor clay	505	179.1	R

*Phases: pl, plagioclase; ol, olivine; px, pyroxene; op, opaque minerals; alt, secondary alteration minerals; chl, chlorite

**Cumulative thickness above sea level

about 8% were rejected. Of the 90 lava units studied in this section, one unit, RF 58, gave meaningless results (not included in Table 2). Virtual geomagnetic pole positions (VGP) were calculated from magnetic declination and inclination values. The azimuth of the dip direction in Holmatindur, section RF, is 270° due west, and the criteria in Table 3 were used to correct the field values for tilt based on lava dips.

Results from sections RD (Kollur) and RE (Teigargerdistindur)

Paleomagnetic results and statistical data for sections RD and RE are presented in Table 2. Units in sections RD and RE were demagnetized using the peak alternating field values: 150, 200, and 250 Oe. If magnetic directions were unstable after demagnetization to 250 Oe, the process was continued to higher values which usually gave stable paleomagnetic directions. Four samples from each unit were demagnetized in sections RD and RE or a total of 253 cores of which all but 7 were considered acceptable after demagnetization. Samples were rejected if stable directions were not attained at 500 Oe. Virtual geomagnetic pole positions (VGP) were calculated after tilt correction, assuming 270° as the azimuth of the dip direction. For tilt correction the following criteria in Table 4 were applied. In the present study, either virtual pole latitudes of lavas are close to the geographic poles with clear normal or reversed polarity, lat.$\geq|40°|$, or they are termed transitional, i.e., if lat.$<|40°|$. Only one lava has a virtual geomagnetic latitude less than $|10°|$. The same criteria have recently been used for the magnetostratigraphy of northern Iceland [*Saemundsson et al.*, 1980]. Especially noteworthy are the many short polarity intervals and lavas of transitional polarity in the Holmatindur section. Thus out of the 90 lava units sampled in Holmatindur in 1979, 16 have transitional polarities (18%). Of these, 12 fall on reversal boundaries on the polarity log of Figure 6.

DISCUSSION OF PALEOMAGNETIC RESULTS

Demagnetization and magnetic intensity.

Several lavas and one dyke unit of the Holmatindur section were selected to display the effects of demagnetization under peak alternating field conditions on (1) magnetic intensity (Figure 3) and (2) magnetic declination and inclination (Figure 4). The selected samples differ with regard to granularity and state of alteration (Table 5).

As can be seen from a comparison of Table 5 and Figure 3, only one of the selected samples, RF 1-p3, the lowest exposed lava and the most altered (mesolite grade), seems to have acquired a soft magnetic component that is removed during demagnetization. With the exception of RF 1-p3, demagnetization has effects on the selected lava samples similar to demagnetization of fresh oceanic rocks [*Fox and Opdyke*, 1973]. The selected rocks tend to be very stable and demagnetization has virtually no effect on magnetic pole positions (VGP) except on lava sample RF 32-p3, which shows a trend toward steeper values with increased applied field strength.

An attempt was made to demonstrate the overall vertical changes of remanent magnetic intensity in the Holmatindur section. For this purpose the Holmatindur section, about 0.9 km thick, was divided into nine lava intervals, each consisting of 10 lava units. For each interval the arithmetic mean of the normal remanent magnetization and remanence after 200 Oe a.f. demagnetization were calculated. The results are plotted in Figure 5 and tabulated in Table 6 together with the standard deviation of the magnetic intensity for each lava interval. Included in Table 6 are comparable values for oceanic pillow basalts [after *Fox and Opdyke*, 1973]. Figure 5 clearly demonstrates two interesting points, namely, (1) the magnetic intensity of the 0.9 km exposed lava succession does not decrease with depth, and (2) the drawn curves for the magnetic intensity before and after demagnetization are parallel but become closer together with depth in the lava pile. The first observation

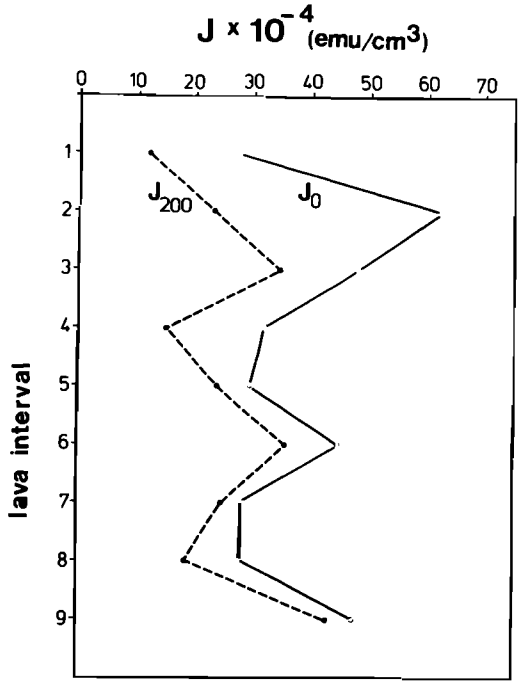


Fig. 5. Magnetic intensity plotted for lava interval. Each value represents a mean average for 10 lava flows. The two curves show magnetic intensity (J x 10^{-4} emu/cm³) before and after demagnetization, i.e., at 0 and 200 Oe, respectively.

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TABLE 6. Magnetic Intensity for Nine Lava Intervals in the Holmatindur Section Compared with Oceanic Pillow Basalts [after Fox and Opdyke, 1973]

Holmatindur, Section RF		J_0 x 10^{-4} emu/cm³		J_{200} x 10^{-4} emu/cm³		$J_0 - J_{200}$	Oceanic Pillow Basalts [Fox and Opdyke, 1973]		J_0 x 10^{-4} emu/cm³
Lava Interval	Units	Arithmetic Mean	Standard Deviation	Arithmetic Mean	Standard Deviation	Percent Decrease	Rock Type	Number of Samples	Arithmetic Mean
1	RF81-90	27.2	19.9	12.1	8.1	55.5	fresh	12	147.8
2	RF71-80	62.6	35.0	23.4	9.6	62.7 } 48.6	slightly weathered	10	37.1 } 31.0
3	RF61-70	48.0	27.8	34.4	21.7	27.6	moder. weathered	18	22.4
4	RF51-60	31.8	32.2	15.3	15.9	52.1 } 30.7	strongly weathered	22	33.4 }
5	RF42-50	29.6	28.7	23.8	21.0	19.4	metamorphosed	10	4.4
6	RF32-41	44.7	57.2	35.6	51.7	20.5 }	all basalts	72	47.6
7	RF22-31	28.2	17.5	24.2	16.3	14.0 } 18.8			
8	RF11-21	27.7	22.5	18.7	19.0	32.6			
9	RF 1-10	47.5	42.8	42.8	45.0	10.0 }			
All lavas	RF 1-90	38.6	34.2	25.6	27.6	32.7			

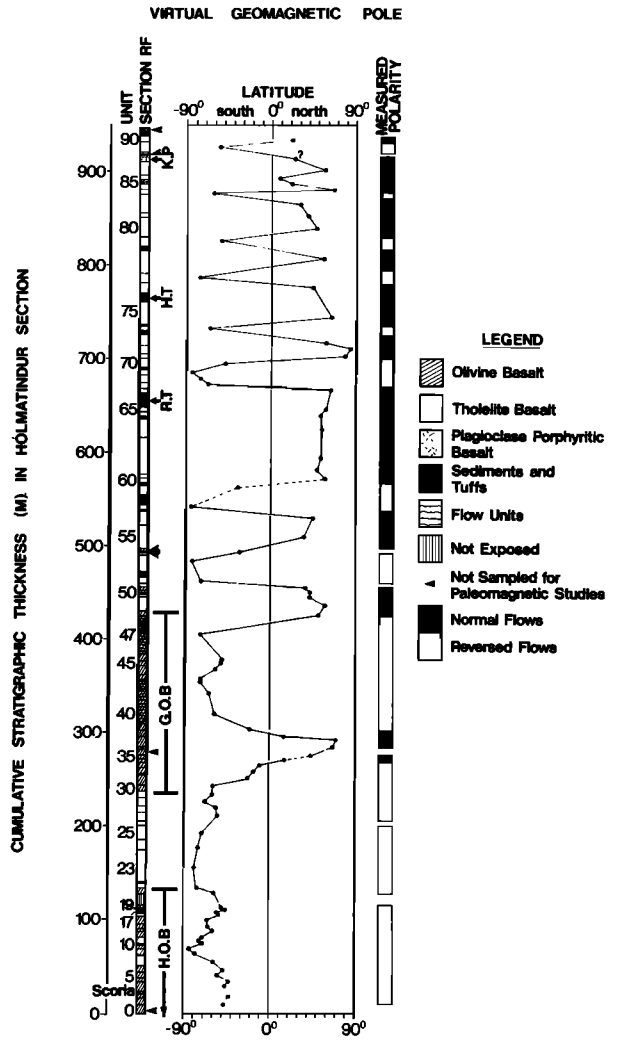


Fig. 6. Polarity log for the Holmatindur section based on latitudes of the virtual geomagnetic pole. The stratigraphic sequences on the left side include the main marker formations (abbreviated).

reflects the heterogeneity of the lava pile with regard to magnetic intensity. The second observation indicates that effects of demagnetization on magnetic intensity decrease with depth so that if the nine lava intervals are divided into three separate groups the decrease in magnetic intensity (J_0-J_{200}) is 48.6, 30.7, and 18.8% respectively from the mountain top to sea level. It is a point of interest that the standard deviation for each of the nine lava intervals both before and after demagnetization (Table 6) is only slightly lower than the arithmetic mean. This clearly reflects the heterogeneity of the overall exposed lava succession, since intralava variations are usually much smaller. In their study, *Fox and Opdyke* [1973] (Table 6) show that weathering of submarine pillow basalts drastically reduces their magnetic intensity. Thus they report a NRM value of 147.8 x 10^{-4} emu/cm³ for fresh rocks but about 31.0 x 10^{-4} emu/cm³ for all weathered rocks and 4.4 x 10^{-4} emu/cm³ for metamorphosed basalts. The Holmatindur lavas have average magnetic intensity values of of 38.6 x 10^{-4} and 25.6 x 10^{-4} emu/cm³ before and after demagnetization, respectively. Both these values are comparable with 31.0 x 10^{-4} emu/cm³ for weathered oceanic pillow basalts. These results are not consistent with earlier work in eastern Iceland, as discussed below.

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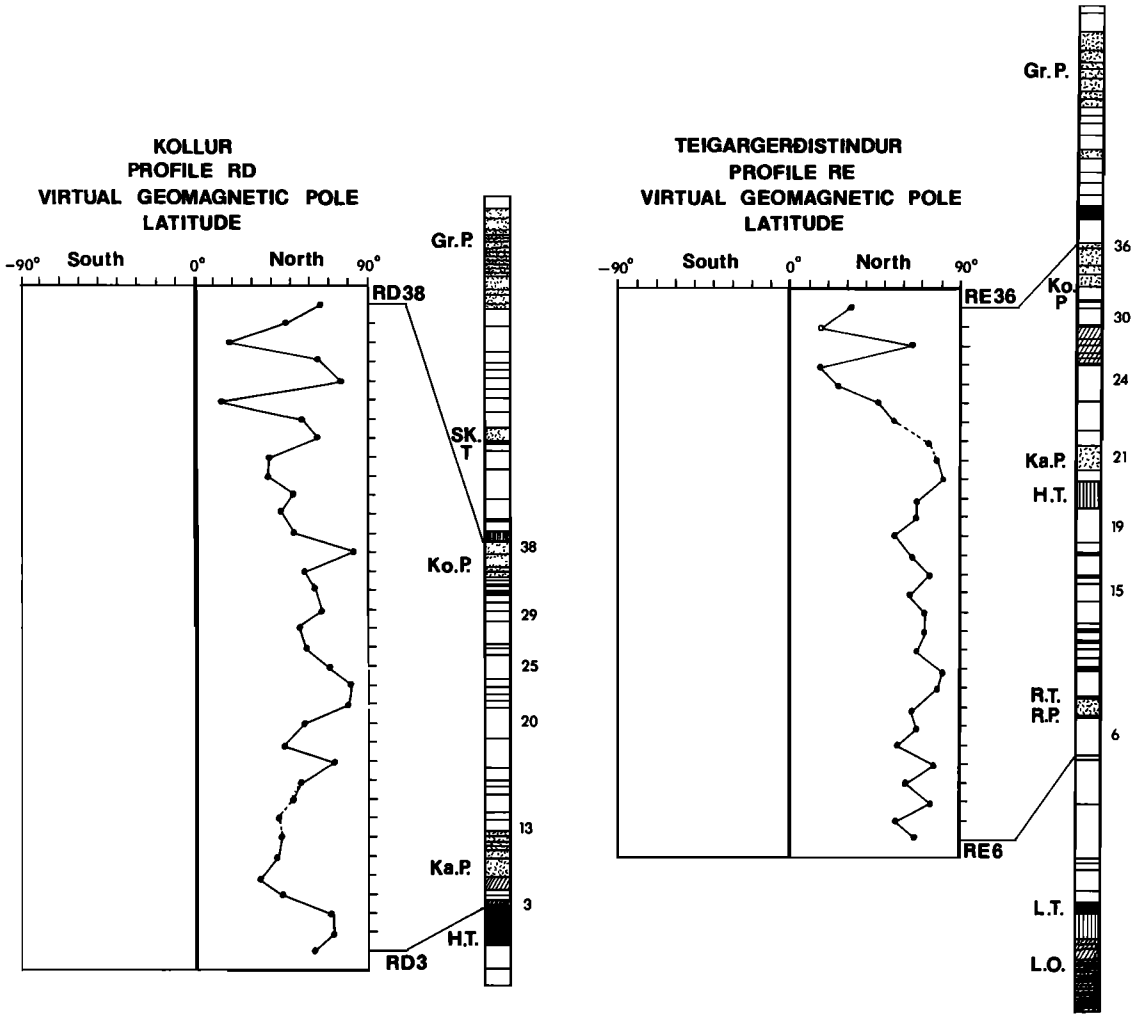


Fig. 7. Polarity logs for profiles RD (Kollur) and RE (Teigargerdistindur) showing latitudes of the virtual geomagnetic pole values. On the right are stratigraphic profiles indicating units that were sampled for paleomagnetic studies.

Watkins and Walker [1977] report how normal remanent magnetization for the exposed basalt lava succession in eastern Iceland decreases linearly with depth below the preerosional plateau surface as a result of increased paleotemperature conditions. They derived this relationship from (1) the work on opaque mineralogy in eastern Iceland by *Ade-Hall et al.* [*1972*] and (2) work by *Walker* [1960] on zeolite grade alteration. They found that for rocks demagnetized under peak alternating field conditions of 100 to 125 Oe, the magnetic intensity decreased nonlinearly with depth and approached a constant value at sea level approximately 1500 m below the preerosional plateau surface.

In summary, the present study does indicate that magnetic intensity of lavas in eastern Iceland decreases neither to the extent nor in the manner reported by *Watkins and Walker* [1977]. If, indeed, magnetic intensity decreases with depth, the heterogeneity of the lava succession is of such magnitude that it masks a systematic decrease with depth. Second, demagnetization affects magnetic intensity in the upper part of the section to a greater extent than it affects lavas close to sea level. This indicates that hydrothermal alteration is more pronounced lower in the lava pile and that the opaque alteration products are magnetically much more stable than the magnetic minerals in the upper part of the lava succession. Here zeolitisation is less pervasive, and only low-temperature zeolites are present (chabazite-thomsonite-stilbite).

Magnetostratigraphy.

Stratigraphic correlations between Holmatindur and the surrounding exposed area proved to be straightforward. Lithological sequences containing the same stratigraphic formations can be traced for tens of kilometers in the strike direction in eastern Iceland, e.g., the Kollur porphyric formation [*Helgason and Zentilli*, this issue].

However, the paleomagnetic signature of these presumably coeval lavas in the investigated sections surrounding Holmatindur [e.g., *Watkins and Walker*, 1977] indicates a totally different polarity history. Here a long, normally magnetized lava sequence corresponds to 'seafloor' magnetic anomaly 5 [*McDougall et al.*, 1976]. The results from the upper part of Holmatindur (Figure 6) indicate eight short, reversed polarity intervals but no long sequence of normally magnetized lavas equivalent to anomaly 5. Thus the frequent short polarity intervals in Holmatindur strongly contradict earlier work in the area that had been coupled with K-Ar dating. Sections RD and RE were sampled in 1980 in order to shed light on this problem. The results from sections RD (Kollur) and RE (Teigargerdistindur) are presented in Figure 7. Here all lavas are normally magnetized. The complete absence of reversely magnetized lavas in both these sections, RD and RE, is in clear disagreement with the Holmatindur lavas. The results from

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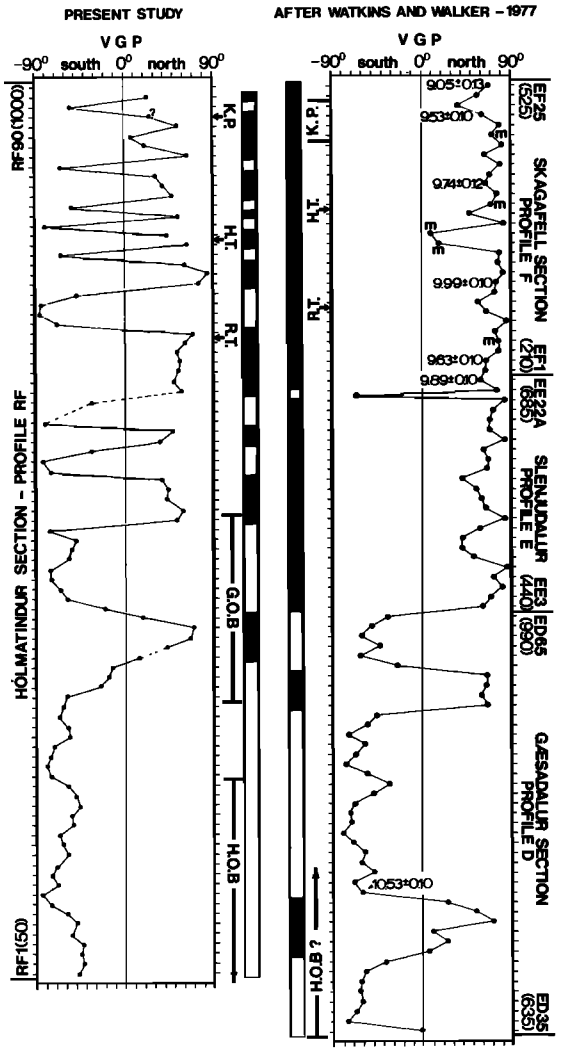


Fig. 8. Paleomagnetic and stratigraphic results of the present study in Holmatindur, section RF, are compared with earlier work on coeval lavas from a different locality, i.e., sections D, E, and F on Figure 4. Units denoted with m are reported values for the second unit sampled in a sequence of thin flow units. Dashed lines indicate that a lava between the plotted values was not sampled. Question marks denote paleomagnetic values that are less reliable. ED 35 denotes profile ED, unit 35; (635) is the altitude above sea level of unit 35. Also shown are absolute age determinations of several samples from *McDougall et al.* [1976].

RD and RE agree, however, with three independent studies in the region. These include paleomagnetic work about 20 km to the north [*Watkins and Walker*, 1977], fluxgate magnetometer results in Teigargerdistindur [*Piper et al.*, 1977], and compass measurements in Holmatindur [*Einarsson*, 1957].

The three sections of this study, RD, RE and RF on Figure 2, are correlated stratigraphically and located perpendicular to the strike direction [*Helgason and Zentilli*, this issue]. From west to east the sections are correlated so that all of section RD is incorporated in section RE updip, although here this interval is represented by fewer lava units. This is due to the generally lower 'tape recording' updip and farther away from a paleovolcanic zone, the extinct ridge axis of which presumably lies buried below sea level west of the IRDP drill hole. Similarly, the entire lava formation of RE is incorporated in RF, but here this lava sequence is higher updip and farther to the east of the assumed paleovolcanic zone. Therefore this lava interval is represented by still fewer lavas in section RF than in RE. As before, the assumption is made that the lavas flowed from west to east; thus lower in the lava pile and closer to the paleovolcanic zone the 'tape recording' was more detailed. From the above it is assumed that if no remagnetization has taken place, all reversals that occur in a high-lying lava section should necessarily also be represented downdip in the lower sections.

Since the paleomagnetic results of the present study show reversely magnetized lavas in the upper part of section RF (but in neither section RD nor RE), it is suggested that secondary magnetization took place in section RF during a reversed polarity interval. This conclusion is supported in the discussion below.

Remagnetization of some 'anomaly 5' lavas.

A closer look at the stratigraphy of the Holmatindur section reveals that the upper lavas, units RF52-RF87, were all sampled in the same gully, which is cut by a 10 m thick dyke of reversed polarity, unit RF 51D (Table 2). No other dyke occurs within 50 m of either side of the gully. Field conditions within the gully did not allow sampling at great distances from the dyke. The distance from the dyke to the first core sampled in a lava was commonly over 10 m, and the minimum was 4 m. This distance was considered ample based on investigations of the effects of intrusives on basaltic host rock [e.g., *Ade-Hall and Dagley*, 1974]. Out of 35 lavas in the gully, 22 are normally magnetized and 13 are reversely magnetized. Five out of eight reversed polarity intervals in the upper part of Holmatindur are each based on a single lava flow of intermediate VGP latitude. Thus these polarity intervals are vaguely defined. Results obtained with portable fluxgate magnetometers have been very useful during initial reconnaissance mapping and stratigraphic correlations in the Tertiary areas of Iceland [e.g., *Saemundsson et al.*, 1980]. However, they are not as reliable as results from cored samples that have been oriented in situ and subsequently demagnetized in the laboratory. In the field season of 1980 fluxgate measurements were conducted within the gully in Holmatindur to help explain the paleomagnetic problem in question. Lavas of 'reversed polarity' were measured at intervals away from the dyke to distances limited by difficult field conditions. A change in polarity toward normal was observed for most of the 'reversed polarity intervals' away from the dyke. It is concluded that the reversely magnetized dyke in the gully may be responsible for the 'reversed events' observed. This would indicate that under certain conditions, remagnetization of lava flows can occur to distances of up to 10 m from a 10 m thick dyke.

Thus, (1) if the reversed polarity intervals in upper Holmatindur are caused by remagnetization and (2) if prior to remagnetization these lavas had normal magnetic polarities, it follows that this lava sequence would correlate perfectly with the magnetostratigraphy of the surrounding area, i.e., normal lava flows of anomaly 5 age. Second, it follows that the reversely magnetized dyke (RF-51D) most likely intruded the upper section after the duration of magnetic anomaly 5, about 9 m.y. ago, i.e., long after the Holmatindur lava pile was formed. An explanation offered for the remagnetization assumes that prior to alteration of the lava pile some of the lavas had low Curie points (perhaps 300°C). Upon intrusion of the dyke, during a reversed polarity interval, the magma temperature was high enough to cause the host lavas to reheat above their Curie points and partly or completely remagnetize some of the host lavas to distances of at least 10 m from the dyke.

This explanation is supported by the work of *Ade-Hall et al.* [1971]. In their study of magnetic properties of lavas from eastern Iceland and elsewhere they found that the least altered lavas with

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no amygdules or those from the chabazite-thomsonite zeolite zones have both low (100-300°C) and/or high (about 500°C) Curie temperatures. This behavior is also displayed by fresh lavas from the neovolcanic zones in Iceland. The bimodal distribution results from local phase splitting of initially homogeneous magnetic minerals into minerals with low and high Curie point temperatures during magma cooling. Conversely, lavas subjected to a higher degree of zeolite alteration, i.e., mesolite-laumontite zones, only have Curie points greater than about 300°C. Here, as a result of hydrothermal alteration, the lower Curie temperatures have been raised. A study of secondary minerals in the upper part of Holmatindur reveals that only minerals of low alteration grade are present. It is therefore assumed that the dyke in the upper part of Holmatindur was emplaced before pervasive alteration could penetrate the surrounding lava pile and raise their Curie temperatures. Thus the thick intruding dyke may have reheated the host lavas above their Curie points, causing remagnetization and a change of magnetic polarity direction from normal to reverse.

CONCLUSIONS

1. Paleomagnetic and lithological correlations of the investigated sections and earlier work in the area clearly establish the presence of 'anomaly 5' age lavas. The lower boundary of anomaly 5 has thus been identified both in the IRDP core [*Bleil et al.*, this issue] as well as in the exposed lava pile east of the drill site.

2. To confirm the remagnetization of lavas in upper Holmatindur, further paleomagnetic sampling would be desirable. The present results may have some implications for the geomagnetic polarity history of eastern Iceland. Thus when the available magnetostratigraphic data for eastern Iceland are examined [*Watkins and Walker*, 1977], some very short polarity intervals are vaguely defined and may possibly be explained by local remagnetization. Thus the reported reversed polarity intervals within anomaly 5 in eastern Iceland (i.e., each of intervals 9a and 9b; *Watkins and Walker* [1977]) are represented by only one lava unit. The postulated remagnetization of upper Holmatindur may primarily be the result of an unusually thick (10 m) basalt dyke intruding through the lava pile. In Reydarfjordur the mean dyke thickness is usually 3-4 m and dykes seldom exceed 10 m. Therefore, extensive remagnetization of lavas is considered unlikely in eastern Iceland.

3. Lava heterogeneity in terms of magnetic intensity of the 0.9 km section in Holmatindur is of such magnitude that no clear decrease is observed with depth. This is in contrast to earlier studies of the exposed lava succession in eastern Iceland. It is suggested that geochemical variations and/or grain size of magnetic phases play an important role in the magnetic record of the lava succession.

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